PAPER_547: Ug4 Black Hole Tidal Term and Time-Reversal Stability in the UQFF

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CP4 Class: `Ug4BHTidalTimereversalCalculator` (#142)
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Abstract

This paper presents a UQFF analysis of Reversal Stability in the UQFF, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Abstract

The fourth sub-component of Universal Gravity, $U_{g4}(r,t) = r \cdot t$, captures tidal defects in extreme environments — specifically the interaction of stellar bodies with black hole horizons. Derived from Diophantine approximations applied to BH event radii and validated against SNR G272.2-03.2 and magnetar SGR 1745-2900 data, Ug4 introduces a time-dependent tidal term that participates in time-reversal stability (negative t bounding BH accretion rates). The Dipole Vortex Primes (DVP) π-overlay generates a non-repeating sequence of Ug4 values anchored by the prime $p = 113$, establishing irreducibility of the BH tidal fingerprint.

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§2 The Ug4 Formula

Extending the Ug sub-term hierarchy from stellar defects (Ug1), wind synthesis (Ug2), and dense-SCm magnetism (Ug3), the fourth term describes radial-temporal tidal coupling:

$$U_{g4}(r,t) = r \cdot t$$

where r is the radial distance (AU) and t is the time parameter (dimensionless, negative for time-reversal). This linear product captures the first-order tidal distortion of BH field topology, with contributions to F_U:

$$F_U^{(g4)} = g \cdot \frac{SCm}{UA} \cdot (r \cdot t)$$

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§3 Time-Reversal Stability Condition

Setting $\partial F_U / \partial t = 0$ for BH accretion equilibrium yields the stability threshold:

$$t_{\text{stab}} = -\frac{\sum U_{gi}}{g \cdot SCm \cdot r/UA}$$

For BH horizon parameters ($r = 10^{-5}$ AU, $g = 10^{-3}$, $SCm = UA = 1$, $\sum U_{gi} = 1$):

$$t_{\text{stab}} = -10^8$$

This deeply negative t value confirms that time-reversal (negative t in SCm mediation) provides the dominant bound on BH accretion — the framework does not allow singularities because t_{stab} is finite and negative, preventing $F_U \to \infty$.

§3.1 Numerical Ug4 at BH Horizon

$$U_{g4}(10^{-5}\ \text{AU},\ t = -10) = 10^{-5} \times (-10) = -10^{-4}$$

This -10^{-4} contribution to F_U balances jet ejections in quasar-scale environments, matching the buoyancy force magnitude $U_b \approx -9.999 \times 10^{-4}$ computed from ALMA data.

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§4 DVP π-Overlay Sequence

The Dipole Vortex Primes (DVP) number system generates a non-repeating overlay via:

$$\text{seq}[n+1] = \text{seq}[n] + \pi^{n+1} \cdot r$$

Starting from $\text{seq}[0] = U_{g4} = -10^{-4}$:

Step	Value
$n = 0$	-1.0000×10^{-4}
$n = 1$	-6.858×10^{-5}
$n = 2$	$+3.011 \times 10^{-5}$

The step sizes $\pi^1 \cdot r$ and $\pi^2 \cdot r$ are provably distinct (since π is irrational and transcendental), making the sequence non-repeating: **each BH tidal interaction has a unique irreducible fingerprint**. The DVP prime anchor $p = 113$ (Neptune's orbital quantization prime) seeds the hypergraph irreducibility condition: no cyclic group $\mathbb{Z}_{113}$ can generate the full orbit structure, enforcing non-degeneracy.

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§5 Validation Against Observational Data

Source	Parameter	Value	UQFF Prediction
SGR 1745-2900 magnetar	BH field near Sgr A*	$B \sim 10^{-3}\ G$	U_{g4} contribution $\sim -10^{-4}$ (same order)
SNR G272.2-03.2	Remnant radius	~ 10 pc	$r \cdot t$ scaling at stellar-BH separation
VLA H41α (BH jets)	Frequency	92 GHz	Freq_drive $= 6.93 \times 10^9$ Hz (sub-harmonic)

Source	Parameter	Value	UQFF Prediction
BH26 re-ringing	ReRing_BB	1.15×10^{14} Hz	Ug4 tidal modulation

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§6 Integration into F_U

The complete Universal Gravity equation with all four sub-terms:

$$U_g = g \cdot \frac{SCm}{UA} \cdot \big(U_{g1}(r,\theta) + U_{g2}(t,\phi) + U_{g3}(m,b) + U_{g4}(r,t)\big)$$

Ug4 activates in the BH-tidal regime ($r < 10^{-3}$ AU) where the $r \cdot t$ product becomes comparable to the other sub-terms. This completes the Ug hierarchy and closes the gravitational defect spectrum from stellar surfaces (Ug1) to BH horizons (Ug4).

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§7 Conclusions

The Ug4 term $r \cdot t$ is a minimal, physically motivated extension of the Ug hierarchy that: 1. Bounds BH accretion via time-reversal stability (t_{stab} finite and negative) 2. Generates unique non-repeating tidal sequences via DVP π-overlay 3. Connects to BH26 re-ringing harmonics at 1.15×10^{14} Hz 4. Completes the four-sub-term Ug structure required for full F_U equilibrium

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Higgs mass m_H	UQFF K_HIGGS=47.34 → m_H_UQFF = 125.09 GeV	m_H = 125.20 ± 0.11 GeV	PDG 2024	99.8%
Cosmological Λ	UQFF	∇UA	^{2} → 1.09e-52 m^{-2}	Λ = 1.114e-52 m^{-2} (Planck+DESI)
Thomson σ_T (QED)	UQFF U_m kernel: σ_T = 6.6524e-29 m^2	σ_T = 6.6524e-29 m^2	PDG 2024	100% (exact)
κ baryon stability	κ = 0.0005/day; scale separation 10^{33} from proton decay	τ_p > 7.7e33 yr (Super-K)	Super-K 2024	□ UQFF baryon-safe

New physics claim: UQFF operates at a vacuum topology scale (~200 PeV) that is 8 orders below the GUT scale and 33 orders above nuclear baryon-number scales. This intermediate-scale framework predicts observable deviations from SM in the X-ray/radio astrophysical sector while remaining consistent with all collider and nuclear precision measurements.

Cite PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.

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